Cognitive Errors in Diagnosis: Instantiation, Classification, and Consequences

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To identify diagnostic errors caused by faulty clinical cognition, we analyzed 40 consecutive transcripts of problem-solving exercises published in a pedagogic series of clinical reasoning. The analysis disclosed multiple errors in cognition and produced a provisional classification of these errors based on a framework derived from cognitive science. Faults in cognition were identified in all steps of the diagnostic process, including triggering, context formulation, information gathering and processing, and verification. We instantiated each type of error by providing detailed specific examples, and identified the consequences of each error. We conclude that cognitive errors can be identified and classified, that they can produce serious morbidity, and that a classification of cognitive errors is a step toward a deeper understanding of the epidemiology, causes, and prevention of diagnostic errors.

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Despite the widely accepted view that accuracy in diagnosis is a critical aspect of patient care, and despite the enormous penalties sometimes paid by both patients and physicians when the diagnostic process goes awry, little effort has been devoted to the description and analysis of diagnostic errors. How do we identify such errors now? We stumble onto them when their consequences lead either to embarrassing or unfortunate patient outcomes. Sometimes we appreciate one has occurred because a certain practice deviates from some normative model (for example, our model may include a routine pelvic examination, and an error is revealed when such an examination is omitted). We also identify errors through the courts: in this arena, an error is considered to have occurred when a physician violates some customary practice pattern.

None of these approaches is satisfactory. Even if they identify many errors in diagnosis, they leave a grossly incomplete picture of the characteristics of these errors: in particular, their nature, frequency, causes, and consequences. Indeed, diagnostic errors are often manifestations of a fallacy in clinical cognition, or clinical reasoning, yet none of these approaches identifies the kind of cognitive error committed. A comprehensive analysis of diagnostic errors would have considerable merit, but it would require an extensive study of physician behavior. Although we cannot offer an exhaustive explication of all the characteristics of diagnostic errors, we can begin to identify and categorize those errors that can be attributed to faulty clinical cognition. In this study, we describe characteristics of diagnostic errors based on an analysis of transcripts of physicians' "thinking aloud" behavior.

SUBJECTS AND METHODS

We identified errors in diagnostic reasoning by analyzing transcripts of problem-solving exercises published as part of a didactic series on clinical problem-solving [1–40].

Rationale for the Methodology

The purpose of the analysis was to identify and categorize diagnostic errors consequent on faulty clinical cognition. We tried to account for the complex behavior that produces diagnostic errors, by first examining available evidence and generating one or more hypotheses that are sufficient to account for the observed behavior [41]. This task is necessarily a process of discovery, i.e., one of hypothesis generation, because no general theory of diagnostic problem-solving exists and because no competing hypotheses about the nature of diagnostic errors have been formulated. The approach we have used is one of instantiation: we know that if we can identify examples of certain kinds of errors, such errors do exist; what this method does not ensure, of course, is that all the possible errors will be uncovered. This experimental approach, drawn from the methods of cognitive science, is designed to produce a classification with prima facie validity. The output of such hypothesis generation is a tentative classification of cognitive errors in diagnosis.

Materials

The material for this study consisted of 40 published reports consisting of patient histories prepared in chronologic segments and transcripts of physicians' "thinking aloud" responses to the clinical material. The clinician discussants were mostly academic internists and surgeons, except for three who were house officers. Prior to publication, the transcripts were edited lightly, sufficient to produce a consistent style for publication, but not so extensively as to change the content of the responses. Cases in this series were not selected by a random process; rather, they were intentionally chosen for publication for their merits in instantiating (i.e., providing specific examples of) various clinical problem-solving tactics and strategies. Most were oriented around problems in diagnosis, although three focused exclusively on therapeutic issues. We selected for study only those 34 cases (contained in 37 of the publications) in which diagnosis was the principal process selected for discussion [1–9,11–19,21–34,36–40]. The published cases and the discussants' responses constitute all the data for this study.

Content of Analysis

Each published transcript was analyzed to assess these attributes: (1) the nature of the clinical disorder, (2) the number of diagnostic errors, (3) the nature of the error, (4) the aspect or segment of the diagnostic process in which the error occurred, and (5) the actual or potential consequences of the error. The details of transcript analysis of "thinking aloud" behavior have been described elsewhere [41–43]. In the present study, extensive transcript evaluation using script analysis, assertional analysis, and referring phrase analysis (as employed in our recent study of medical management decision-making [41]) was unnecessary because we intended only to identify and categorize errors, not to investigate their pathogenesis. Indeed, the errors we sought generally can be unambiguously identified from the transcripts. Because all the data used in this study are already published [1–40], the reader can carry out an independent check on the validity of our assertions concerning the existence and types of such cognitive errors.

A Framework for the Results

A pragmatic framework must be sought upon which to assemble a cohesive picture of diagnostic errors. Because the few studies and reviews of diagnostic errors provide no such structure, we have proposed one. Logically, any such structure should have as its basis the cognitive outline of the diagnostic process. Unfortunately, this outline remains incomplete and current workers in the field would quite likely produce different versions. Rather than trying to invent this framework, we will use an outline adapted from problem-solving in non-medical domains: one that is based on modern cognitive science and that follows the process of scientific discovery [44]. The steps in this process are as follows:

(1) Triggering one or more hypotheses: Triggering is the process of establishing one or more diagnostic hypotheses from information about a patient. Typically, this information consists of the patient's age, sex, race, and presenting complaints. We use the term "triggering" here to describe the evocation or generation of diagnostic hypotheses because this terminology has been widely adopted in research in artificial intelligence and cognitive science [45–49]. Both we and others, however, have described triggering as hypothesis activation, hypothesis generation [50–52], or hypothesis evocation [53].

(2) Developing a cognitive representation (context) of the problem: Framing a problem, or developing a cognitive representation of the patient's problem, involves establishing a context within which the problem presumably will be solved [52]. The context, typically a disease, a syndrome, or a clinical entity, provides an important constraint on the searching needed to focus in on a specific diagnosis. The concept of context [52,53] appears to be identical to notions of "problem space" [44,49,54,55] or the "logical competitor set" described by others [46].

(3) Gathering and processing information: This procedure involves interpreting existing data, collecting additional relevant data, and developing and revising a working hypothesis [46,50–52,56]. It is carried out by assessing disease prevalence, combining data associated by probabilities and/or causal relations [57], and in some instances applying clinical axioms.

(4) Verification: Verification of a diagnosis is the final step in the diagnostic process. It involves assembling a "final" diagnosis that is highly likely, ascertaining whether causal relations are appropriate, assessing whether all the patient's manifestations are explained by the "final" diagnosis, and ensuring that other reasonable diagnostic possibilities have been excluded [51,52]. Acceptance of a diagnosis is a matter of belief: some physicians accept a diagnosis as verified with a certain amount of evidence, whereas others might demand more evidence, including more attempts to disprove alternative diagnoses.

After all errors were categorized according to the aforementioned schema, several others came to attention. These errors produced flawed diagnoses, but the errors could not reasonably be attributed to faulty cognition, and they were designated as "no-fault" errors. Some of these errors occurred because the patient's clinical findings were highly atypical for the disease they harbored; in other instances, the patient's disease was evolving acutely. In still other instances, the disease was extremely uncommon, and the diagnosis quite likely would be missed by most physicians. This concept of assessing a judgment on the basis of the frequency with which it occurs in the natural environment is not new, and is known as a "prevalence-sensitive" measure [58].

The classification of errors derived from the aforementioned cognitive framework is given in **Table I.** It should be noted that although identification of errors in these steps of the diagnostic process is rather unambiguous, the assignment of such errors to each step is less precise because of overlap between some of the steps in the process. Thus, although another observer might categorize some of the errors differently, this lack of precision does not alter the results, because the goal of the study was to identify the types of errors, not to count them.

RESULTS

General Description of Errors

Table II presents the "final" diagnosis of the 34 patients with diagnostic problems. A wide spectrum of clinical material is represented in these cases: most are medical disorders, and a few are surgical disorders. Many are rare diseases, but a large number are common conditions. Errors were attributed both to the physicians responsible for the patient's care and to the physicians discussing the patients' clinical material during a "pencil and paper" exercise.

Because of the process by which cases were selected for publication (i.e., for their explication of problem-solving tactics), the published transcripts yield a rich source of errors: 42 errors were identified among 20 of the published cases (Table II). In 14 of the published cases, no error could be identified. When a diagnostic error was identified, an attempt was made to determine whether or not the error could be attributed to fallacious reasoning. As noted before, those that could not were designated as "no-fault" errors.

The identified errors described in the next paragraphs follow the outline shown in Table I. Note that errors are only identified and categorized: no attempt was made to quantify them in each category because the method by which case material was selected precludes describing the epidemiology of errors. Because identification of cognitive errors is so critically dependent on instantiation, we provide detailed examples of each error.

TABLE I

Classification of Cognitive Diagnostic Errors

Faulty triggering
Faulty context formulation
Faulty information gathering and processing
  Faulty estimation of disease prevalence
  Faulty interpretation of a test result
  A faulty causal model
  Overreliance on a clinical axiom
Faulty verification
"No-fault" errors

The Specific Errors

TRIGGERING ERRORS. *A detailed example of defective triggering* [33]: Prior to giving a patient intrave-

TABLE II

Specific Diagnostic Errors

Case	Diagnosis	Reference	Error	Faulty Triggering	Faulty Context	Faulty Information Gathering and Processing: Faulty Estimation of Prevalence	Faulty Data Interpretation	Faulty Causal Model	Overreliance on Axioms	Faulty Verification	No Fault
1	Ethylene glycol poisoning	[1]	No								
2	Periodic paralysis with hyperthyroidism	[2]	No								
3	VIP-secreting tumor	[3]	Yes	x							x
4	Insulin resistance	[4]	No								
5	*Streptococcus bovis* endocarditis	[5]	Yes	x		x		x			
6	Insulinoma	[6]	Yes							x	
7	Left atrial myxoma	[7]	Yes	x							x
8	Masked hyperthyroidism	[8]	Yes	x	x						
9	Ulcerative colitis complicated by *Clostridium difficile* infection	[9,36]	Yes	x							
10	Appendicitis	[11]	Yes				x				x
11	Gastritis	[12]	Yes			x	x				
12	Syndrome of inappropriate ADH secretion	[13]	No								
13	Pulmonary emboli	[14]	No								
14	Systemic atheroembolism	[15]	Yes	x			x	x		x	
15	Cavernous hemangioma of liver	[16]	Yes						x		x
16	Anticoagulant-related renal hematoma	[17,24]	Yes		x	x	x				
17	Coronary artery disease	[18]	No								
18	Hepatorenal syndrome	[19]	No								
19	Myoglobinuria	[21]	Yes	x	x		x			x	
20	Catheter-related sepsis in an immunosuppressed patient	[22]	Yes				x				
21	Complete heart block	[23]	Yes	x						x	x
22	Water intoxication	[25]	No								
23	Lung cancer	[26]	No								
24	Hypertension secondary to spinal cord disease and urinary tract obstruction	[27]	Yes			x	x				
25	Acute *Yersinia* colitis	[28,37]	No								
26	Malabsorption syndrome	[29]	No								
27	Depression	[30]	Yes		x	x					
28	Magnesium intoxication	[31]	No								
29	Perforated peptic ulcer	[32]	Yes		x				x		x
30	Drug-induced hypertension	[33]	Yes	x							
31	Drug-induced cholangitis	[34]	No								
32	Abdominal angiosarcoma	[38]	Yes							x	
33	Drug-induced methemoglobinemia	[39]	Yes	x							
34	Adult-onset Still's disease	[40]	No								

VIP = vasoactive intestinal peptide; ADH = antidiuretic hormone.

nous L-dopa as part of an evaluation for possible hypopituitarism, a physician failed to obtain an adequate medication history (the patient was receiving an antidepressant medication). Shortly after the infusion, the patient became acutely and severely hypertensive. We presented the clinical data to two physicians on separate occasions. One (a cardiologist) immediately and correctly inferred that the patient was receiving a monamine oxidase inhibitor, whereas the second physician (an endocrinologist) never mentioned the correct diagnosis and instead raised possibilities of pheochromocytoma and multiple endocrine neoplasia. After the second physician was told the correct diagnosis, he acknowledged that he knew about the entity but had not thought of it. We interpret the error on the part of the endocrinologist as a triggering error.

Other triggering errors: In one instance, the physicians caring for a patient seemed unaware of a diagnostic hypothesis (atheroembolism) and thus the clinical findings in the patient did not trigger this diagnosis [15]. In two instances (atrial myxoma [7] and drug-induced myoglobinuria [21]), the clinicians were aware of the diagnostic entity, but failed to appreciate that the patient's clinical features should have pointed to this diagnosis.

ERRORS IN CONTEXT FORMULATION. *A detailed example of faulty context formulation* [8]: A 53-year-old woman had a three-month history of abdominal pain, anorexia, a 10-kg weight loss, and vomiting. Physical examination disclosed a wide pulse pressure, brisk reflexes, and palmar erythema. Her thyroid gland was not palpable. Results of laboratory studies were normal except for mild hypercalcemia. Extensive evaluation of her gastrointestinal tract by her physician was unrevealing. The correct diagnosis, hyperthyroidism, was made only after she developed a brief episode of atrial flutter-fibrillation during a hospitalization for diagnostic tests. The gastroenterologist to whom we presented the clinical data raised the correct diagnosis with far less data than were available to the patient's physician. We interpret the rather extensive gastrointestinal workup by the patient's physician as a framing error based on the faulty assumption of a gastrointestinal disorder as a context for her illness.

Other errors in context formulation: In one example, a physician identified a specific context, namely a mass lesion in the urinary tract, but so closely equated the mass with neoplasm (incorrectly: it was a hematoma) that the correct diagnosis never emerged [17,24]. In another framing error, the physicians considered gastrointestinal bleeding in a patient and staunchly held to this context even when the patient developed considerable evidence for perforation of a viscus; the perforation was discovered accidentally when free air was seen on an upright chest radiograph obtained because the patient developed a productive cough [32].

ERRORS IN GATHERING AND PROCESSING INFORMATION. (1) ERRORS IN ASSESSMENT OF DISEASE PREVALENCE. *A detailed example of faulty assessment of prevalence* [27]: On two occasions, a 38-year-old man with a congenital lumbosacral myelomeningocele developed severe hypertension concomitant with an episode of urinary tract obstruction. After the second episode, a physician who was called to see the patient in consultation performed an intravenous phentolamine test, the result of which he interpreted as positive. Multiple catecholamine determinations yielded some abnormal and some normal results. Still later, multiple studies failed to confirm the diagnosis of pheochromocytoma. When we presented these clinical findings to a nephrologist, he immediately made the correct diagnosis and was not persuaded that the patient might have a pheochromocytoma even when he was given all of the positive test results. We interpret the original consultant's error as one related to the representativeness heuristic [59]. Presumably, the consultant considered the diagnosis of pheochromocytoma quite likely given the representative finding of an abrupt severe elevation of blood pressure and the response to phentolamine. He disregarded the extremely low prevalence of pheochromocytoma, while apparently thinking little of the far more prevalent combination of spinal cord disease and urinary tract obstruction as the cause in such a patient.

Other defects in assessment of prevalence: In other instances, clinicians incorrectly assumed that an uncommon disease was quite likely because the patients' clinical findings resembled the clinical findings of an uncommon disease (Zollinger-Ellison syndrome [12] and renal cancer [17]). This error also is attributable to the representativeness heuristic [59] and is quite the opposite of the aforementioned triggering errors: In those triggering errors, rare diseases were not invoked (atrial myxoma; myoglobinuria) even when clinical features were consistent, whereas rare diseases were inappropriately considered likely in these instances.

Another error attributable to inappropriate assessment of prevalence involved an elderly man with sweating episodes and weight loss. Although his physician suspected that the patient was depressed, he began investigating the patient for a tumor secreting a vasoactive substance when he heard an eidetic description of the patient during a sweating episode. This exaggerated view of prevalence of a secretory tumor presumably was based on the availability heuristic [30,60].

(2) ERRORS IN INTERPRETATION OF CLINICAL DATA. *A detailed example of faulty interpretation of clinical data* [21]: A 40-year-old man, seen in the emergency room for evaluation of severe back pain, was found to have renal insufficiency (serum creatinine level 3.7 mg/dl). He was known to abuse alcohol and other drugs. Examination showed no abnormalities except for paraspinal and costovertebral angle tenderness, and his urine was the color of cola soda. Urine "dipstick" demonstrated strongly positive findings for blood, but the urine sediment contained no red cells. The physicians taking care of him and a nephrologist to whom we presented his clinical findings quickly made the diagnosis of myoglobinuria-induced renal failure, but another physician who received the same data did not. He considered renal damage from a contaminant in an intravenous injection, an anatomic abnormality of the urinary tract, liver disease, and acute nephritis. Even after he learned that the patient's creatine phosphokinase level was elevated, he failed to make the connection between rhabdomyolysis and renal disease. We consider this error a fault in incorporating and properly interpreting clinical data. (Note that this error also was designated a triggering error.)

Other defects in interpretation of data: In another instance, a positive test result (high plasma gastrin level) was interpreted as support for a disease entity (Zollinger-Ellison syndrome) rather than as a false-positive value [12]. This error, failure to appreciate that a positive test result in a population with a low disease prevalence is quite likely to be false-positive (unless the test has an exceptionally high specificity), is frequently made by physicians [61,62].

A second error in test result interpretation was one in which inordinate weight was given to the findings of a single test. In one observed instance, an arteriogram showing abnormal vessels and a "tumor blush" was weighted far more strongly than several other studies that favored a neoplasm considerably less; the mass, as described earlier, was later found to be an anticoagulant-related hematoma [17,24]. In the same case, the language used by physicians may have contributed to the error in interpretation of the laboratory test results. Terms used by the radiologists such as "mass" and "tumor" probably prejudiced the clinicians toward believing that a neoplasm was present; the term "space-occupying lesion" probably would have produced less bias.

(3) ERRORS IN CAUSAL REASONING. *A detailed example of faulty causal reasoning* [15]: A 61-year-old woman with severe vascular disease developed hypertension, hematuria, impaired renal function, confusion, and purplish discoloration of her hands and toes several days after a cardiac catheterization. Her physicians carried out extensive immunologic studies, performed renal arteriography in a search for evidence of vasculitis, and treated the patient with corticosteroids and cyclophosphamide. A nephrologist to whom we presented the patient's clinical data promptly made the correct diagnosis of catheter-induced atheroembolism. In this instance, the patient's physicians correctly deduced that the disorder involved the small arterioles, but they failed to appreciate that the lesion was vascular obstruction by cholesterol crystals. The causal attribution to inflammation not only was incorrect, but led to inappropriate tests and treatment.

Other defects in causal reasoning: In another case, an inappropriate notion of causality led to an incorrect diagnosis. In this instance, a patient with known valvular heart disease presented with severe congestive heart failure. The clinicians taking care of the patient failed to consider endocarditis as a possible cause of the patient's declining cardiac status [5].

(4) ERRORS IN APPLICATIONS OF CLINICAL AXIOMS. *A detailed example of faulty application of a clinical rule* [16]: A 35-year-old woman presented to the hospital with diarrhea and right upper quadrant abdominal pain. Examination disclosed guarding but no rebound tenderness in that region, and the only abnormal laboratory result was a slightly low hematocrit. An ultrasound study of the right upper quadrant disclosed multiple hypoechoic masses that appeared to be confluent, and diagnoses of possible liver abscess or metastatic tumor were raised. Both the patient's physicians and the physician to whom we presented the patient's findings considered performing needle aspiration biopsy of the liver mass. The discussant invoked "Sutton's law" to explain why such a procedure was warranted. Fortunately, a computed tomographic scan identified the lesion as a cavernous hemangioma, and no further studies were carried out. We interpret this potential error, namely the choice of liver biopsy, as an example of faulty application of a common but fallible clinical rule.

Other defects in application of axioms: In another instance, clinicians failed to diagnose a perforated peptic ulcer in a patient who presented with gastrointestinal bleeding, perhaps because they may have been relying excessively on the clinical axiom that "bleeding ulcers don't hurt and hurting ulcers don't bleed" [32].

ERRORS IN VERIFICATION. *A detailed example of faulty verification* [38]: A 44-year-old normotensive man was found to have a large adrenal mass adherent to the liver at laparotomy. Pathologists described the mass grossly and microscopically as a pheochromocytoma. As in one of the other cases that we have described, results of some of the catecholamine studies were abnormal and some were normal. Thereafter, recurrent bleeding into the peritoneal cavity occurred, requiring two additional laparotomies. Despite many atypical findings, the patient's physicians and the internist to whom we presented the clinical data accepted the diagnosis of pheochromocytoma, but an oncologist who saw the patient in consultation thought that recurrent intraperitoneal bleeding was so inconsistent with this diagnosis that he disputed the pathologists' opinions and asked for further special histologic studies of the tumor. These studies documented that the tumor was not a pheochromocytoma, but an angiosarcoma. We consider this error an example of premature closure resulting from failure to recognize that the diagnosis of pheochromocytoma did not explain the findings; the error is thus a failure of verification.

Other examples of faulty verification: We observed several other examples of faulty verification. In one instance, no clear diagnosis was achieved because the physician not only framed the problem incorrectly but missed a discriminating diagnostic clue, and thus he failed to assemble a coherent synthesis of the patient's clinical findings [21]. In another instance, the diagnosis accepted by a clinician as correct (vasculitis) failed to explain all the clinical findings, yet the diagnosis was not questioned [15]. Inappropriate adherence to a diagnosis despite convincing evidence for another diagnosis was the cause of error in another instance. In this example of "premature closure" [63], evidence mounted for a diagnosis of insulinoma in a young woman with diabetes in her family, but one physician continued to maintain that the patient had exogenous hyperinsulinism even when all the test results pointed to the diagnosis of insulinoma [6].

"NO-FAULT" ERRORS. *A detailed example of a "no-fault" error in diagnosis* [3]: A 53-year-old woman with persistent diarrhea later found to have a metastatic tumor of the liver secreting vasoactive intestinal peptide was evaluated for the diarrhea after she returned from a foreign trip where she had eaten in restaurants not recommended for health reasons. Initially, her diarrhea was moderate. Two parasites were found in her stool, a diagnosis of parasitic diarrhea was made, and she was treated for this disorder. However, diarrhea worsened. The correct diagnosis was made when the patient's condition failed to respond to treatment, diarrhea became massive, and the patient developed refractory electrolyte disturbances. The initial error in diagnosis can be attributed to the rarity of this

TABLE III

Adverse Consequences of Diagnostic Errors

	Reference
Actual Consequences	
Delayed chemotherapy for cancer	[3]
Unnecessary tests of gastrointestinal tract	[8]
Delay in treatment of perforated appendix	[11]
Inappropriate treatment with corticosteroids and immunosuppressive agents; unnecessary and potentially dangerous renal arteriography	[15]
Unnecessary loss of a kidney	[17,24]
Delay in removal of infected intravenous catheter	[22]
Repeated syncopal episodes	[23]
Anxiety about the possibility of a tumor; unnecessary testing for pheochromocytoma	[27]
Unnecessary tests of gastrointestinal tract; delay in treatment for depression	[30]
Delay in treatment of perforated ulcer	[32]
Delayed chemotherapy for cancer	[38]
Potential Consequences*	
Embolic event or pulmonary edema	[7]
Unnecessary tests of gastrointestinal tract	[12]
Bleeding from percutaneous puncture of hemangioma	[16]
Unnecessary loss of colon	[36]
Prolonged hypoxemia	[39]

* Consequences were considered possible, or "potential," if the patient's outcome had been determined by judgments or decisions made by either the patient's physician or the discussant in the published report.

clinical entity, to its nonspecific early manifestations, and to the emergence, presumably by chance, of a reasonable alternate (but incorrect) diagnosis of parasitic disease. None of these factors are errors in cognition, and therefore we identify the error as "no-fault."

Other no-fault errors: We observed several other "no-fault" errors. For example, we identified two examples of "no-fault" triggering. In one of these cases, many physicians did not make the correct diagnosis (complete heart block) for more than 15 years [23], but because the patient's symptoms were quite atypical, designating this fault in triggering as a cognitive error seems inappropriate. In two other instances, the disease itself probably was evolving rapidly, thus producing atypical clinical manifestations. In these two instances, a retrocecal appendix and a peptic ulcer were in the process of perforating; manifestations of a perforated viscus became progressively more typical over time [11,32].

Prevalence of Disease versus Type of Error

As noted before, the method of case selection makes any statement about error prevalence invalid. However, it is worth commenting about the relation in our cases between the prevalence of the disease or condition and the type of error observed. When the relative prevalence of disease was assessed on a four-point scale (1 = common; 4 = rare), we found that triggering errors and verification errors were frequent among the rare diseases and were infrequent among the common diseases. No such consistent pattern was observed for the other types of errors.

Consequences of Error (Table III)

The consequences of these errors are readily identifiable. In the young woman with the high gastrin level, in the man with hypertension incorrectly thought to have a pheochromocytoma, and in the elderly man with sweating episodes, the consequences were anxiety over the possibility of a secretory tumor, and expense of unnecessary testing [12,27,30]. In the patient with atheroembolism thought to have vasculitis, the cost was exposure to ineffective and hazardous drugs, namely corticosteroids and cyclophosphamide [15]. The anticoagulant-treated woman whose kidney was removed for a tumor that turned out to be a hematoma had unnecessary loss of a kidney and costly surgery [17,24]. The woman with heart block was thought for years to have epilepsy or hyperventilation attacks, and eventually might even have had a cardiac arrest if the correct diagnosis had not emerged [23]. One patient might have died of peritonitis if a chest radiograph had not disclosed air under the diaphragm as an "incidental" finding [32]. The patient with the abdominal angiosarcoma might have been spared an unnecessary laparotomy if the correct diagnosis had been made sooner [38].

COMMENTS

Our study reveals striking examples of many cognitive diagnostic errors that led to morbid consequences. If diseases can be considered errors in normal structure and function, and if diseases can be classified by type, etiology, pathogenesis, epidemiology, prevention, and treatment, then, by analogy, diagnostic errors can be considered fallacies in normal clinical reasoning and such errors can be organized into a similar classification. In our study, we identify only the type of errors found in the various segments of the diagnostic process. We claim only that such types of errors exist: we acknowledge that one swallow does not make a summer, yet finding one swallow at least proves that swallows exist. We offer a classification based on elements in the process of diagnosis. The formulation of a suitable classification of errors should promote further investigation into the epidemiology, etiology, pathogenesis, and prevention of diagnostic errors.

The Nature of Cognitive Errors

It is reasonable to assume that faults in clinical cognition that provoke diagnostic errors are the consequences of inadequate knowledge, defective information processing, or some combination of the two. Although we have little data from our study that bear on the relation between the structure or adequacy of physicians' knowledge and the commission of errors, some information on the interplay between cognitive processes and knowledge is available. In one series of studies, defective triggering was attributed to improper interpretation of clinical cues, failure of properly identified cues to raise the possibility of a given disease, or lack of knowledge to invoke a disease [46,64]. Another group of errors was attributed to a fault in physicians' disease model (faulty context formulation in our classification). In another class of errors, a correct diagnosis was eliminated when the clinical findings actually were consistent with this diagnosis [46,64]. This error was attributed to the clinicians' having overly specific expectations for the disease. The same investigators also showed that certain persons moved from one diagnostic hypothesis to another when strong cues were provided: this phenomenon, also described in our study, appears to have its roots in the availability heuristic (see next paragraph) [30,60]. Finally, they showed that physicians sometimes failed to recognize that observed findings were at odds with those of the suspected disease (failure of verification in our classification). The authors attributed this error to an overestimation of the allowable range of variation for findings in a given disease [46,64]. From our study we can say little about the causes of the errors we observed, but it was interesting that triggering errors and verification errors were frequent among the rare diseases and infrequent among the common diseases, suggesting, as might be anticipated, that the prevalence of diseases influences certain clinical cognitive functions.

Cognitive Biases in the Laboratory

In a further effort to understand the nature of cognitive errors in diagnosis, we can assess whether cognitive biases identified in the psychology laboratory are relevant to clinical cognition. First, we describe the results of such studies briefly. Many studies in psychology have shown that people make errors in information processing when solving commonplace problems [65]. Several investigators have used simple problems as their experimental models and predominantly non-physicians as their subjects. The studies have identified several of the important heuristics (rules of thumb) that persons use in solving everyday problems and the errors that persons make when using these heuristics. Three of the principal heuristics are called "representativeness," "availability," and "anchoring." The "representativeness" heuristic is a technique used in probability assessment [59]. It derives from the practice of assessing the likelihood of an event on the basis of its close resemblance to other well-defined events. In one classic experiment, this error was revealed by describing the personal attributes of an introverted and meticulous individual and then asking persons whether they thought he was most likely an engineer, a physician, an airline pilot, or a librarian. Indeed, the persons were confident that the individual was a librarian even if the description was scant, unreliable, or outdated. The "availability" heuristic involves assessing the chance of some event or outcome on the basis of readily recallable similar events or outcomes [60]. The event or outcome may be particularly easy to recall because a given event was quite striking or impressive, because a combination of findings brings it readily to mind, or because the causal connections between events make a given outcome quite imaginable. In a classic experiment that revealed this error, people were asked to judge how many persons on a list were males and how many were females (half were of each sex). Manipulating the list to contain either a disproportionate number of famous males or famous females made people guess that the numbers were not evenly split between the sexes. Another heuristic identified in these psychologic studies is that of "anchoring" [65]. This approach involves assessing the likelihood of an event or an outcome based on some starting point or some initial value. In another classic experiment, one group of persons was asked to estimate the product of $8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$, and another group was asked to estimate the product of $1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8$. The median score of the former group was 2,250, and of the latter group, 512. (The correct answer is 40,320.)

Consequences of Cognitive Biases

Do cognitive biases taint everyday clinical reasoning? Do they influence clinical outcomes? Indeed, physicians make errors similar to those described by the psychologists. In a study we carried out some years ago, we found that physicians presented with a problem of interpreting a result of a diagnostic test (which they agreed was similar to tests in their everyday practice) made grossly incorrect interpretations of a hypothetical positive test result for cancer because they ignored the base rate of cancer in the population [61]. This error is identical to that which we observed in the aforementioned patient with the elevated plasma gastrin level. Several studies in the psychology literature have shown that such cognitive biases also occur in medicine [66–69]. Our study provides further evidence to support this notion. Among our cases, we identified several errors in the use of the representativeness heuristic, and one error in the use of the availability heuristic [12,17,27]. Although we did not identify an error attributable to the anchoring heuristic, other studies clearly show that physicians do make such errors [69].

Thus, cognitive errors exist, yet some investigators have quite appropriately questioned whether such errors are important in real-life problem-solving [58]. They point out that, by and large, people are excellent problem-solvers, that the laboratory exercises may not represent real-world problem-solving, and that laboratory experiments have largely failed to consider the outcomes of biased problem-solving. They argue that the real world often consists of redundant cues and multiple measures of the same cue, and the context in actual problem-solving is likely to be far richer in content than that of the artificial constraints of a laboratory experiment [58]. Indeed, in one laboratory study in a medical domain, physicians faltered in estimating the probability of pneumonia from clinical data and they erred in using data from chest radiographs ordered to diagnose pneumonia [70,71], yet the actual performance of these physicians in diagnosing pneumonia was affected little by these errors [58]. These results have been interpreted as indicating that the cognitive biases observed in experiments reflect more on the inadequacy of the experimental design and normative theory being used to evaluate human problem-solving than they do on the inadequacies of humans as problem-solvers. These opinions notwithstanding, we believe that our study not only demonstrates the existence of these cognitive biases in day-to-day medical decision-making, but also documents the gravity of such errors. Serious emotional consequences and many morbid outcomes resulted from faulty reasoning. Although we did not observe a fatal consequence, several of the scenarios could have been fatal, as we described in detail.

Of course, several types of errors exist besides those that we identified. In this study, we attempted only to identify and categorize errors that could be attributed to faulty cognition. We did not include erroneous diagnoses caused by factors such as ego bias [66], hindsight bias [72], or regret [73,74]. These sources of error may be as important as the errors on which we focused, but they seem to be related more to extraneous factors than to fallacious cognition. Finally, we did not observe errors in which the probability of a given diagnosis and the utility of making that diagnosis were not considered separately, although we also know that such errors exist [75]. In such instances, the probability of the diagnosis may be magnified considerably, although inappropriately, because the outcome associated with a given choice is particularly unfavorable. This error is a kind of value-induced bias [74].

Significance

The study reported herein has produced a classification of cognitive errors in diagnosis, and provides instantiation (specific clinical instances) of such type of error. Some of the errors yielded anxiety, morbidity, and cost, and some threatened life. Rather than taking an exclusively legalistic view of errors in diagnosis as an index of potential liability, or an exclusively bureaucratic view to satisfy one or more boards of accreditation, we also should be taking a cognitive view, aimed toward understanding the etiology, pathogenesis, and epidemiology of such errors. A deeper understanding of cognitive errors is essential in the tasks of teaching clinical reasoning [76–78] and avoiding tragic diagnostic mistakes.

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